Division – Node.js application with Jest tests

Cloud Services · Programming assignment

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Repository https://github.com/AkseliPolo/division

The application

A simple Node.js project that exports a `divide(a, b)` function in `src/calculator.js`. Before dividing, the function validates its input and throws `TypeError` if either argument is not a number or is `NaN`, and `RangeError` if the divisor is zero. `src/index.js` runs the function and prints the result. Tests are written with Jest and run with `npm test`.

Tests

`__tests__/calculator.test.js` contains one positive test case and five invalid-input cases:

#	Test	Expected result
1	divides two valid numbers correctly	`divide(10, 2)` returns `5`
2	first argument is not a number	`TypeError`: Both arguments must be numbers
3	second argument is not a number	`TypeError`: Both arguments must be numbers
4	an argument is `NaN`	`TypeError`: Arguments cannot be NaN
5	divisor is zero	`RangeError`: Division by zero is not allowed
6	arguments are missing	`TypeError`: Both arguments must be numbers

Running the tests

```
> division@1.0.0 test
> node --experimental-vm-modules node_modules/jest/bin/jest.js

(node:28036) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
 PASS  __tests__/calculator.test.js
  divide
    √ divides two valid numbers correctly (2 ms)
    √ throws TypeError when the first argument is not a number (5 ms)
    √ throws TypeError when the second argument is not a number (1 ms)
    √ throws TypeError when an argument is NaN (1 ms)
    √ throws RangeError when dividing by zero (1 ms)
    √ throws TypeError when arguments are missing (1 ms)

Test Suites: 1 passed, 1 total
Tests:       6 passed, 6 total
Snapshots:   0 total
Time:        0.133 s, estimated 1 s
Ran all test suites.
PS C:\Users\aksel\Desktop\Cloude_Services\division>
```

Output of `npm test` in the project directory – 1 test suite and 6 tests passed.